# Learning Local Coloring Rules: Continuation Fidelity, Progress, and Failed Repair

Research collaboration
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**Abstract**

Can perfect agreement on a teacher's observed actions conceal an incorrect local rule? In a distributed coloring framework, three BE/E1 training procedures agree on every measured native teacher action, yet disagree sharply when the proposed color or participation message changes. We define these typed continuations explicitly and audit both occurrence-weighted accuracy and sufficient local input types. The historical counterfactual procedure is more accurate, but uses more labels and mostly familiar types; a separate forty-model, budget-matched experiment tests held-out input types. GM improves from 50% to 100% balanced accuracy when interventions supply negative color confirmations absent from the native control; autonomous success is unchanged. BE has an uncertain mean gain and fewer autonomous successes (7,405 versus 9,204 of 10,000). A post-test eligibility-only diagnostic scores 99.71% BA versus CF's 90.41%, whereas CF has higher equal-type accuracy (86.94% versus 81.27%). The held-out input contexts mostly reuse logical oracle configurations seen in training. Frozen transfer exposes a complementary limitation: correct rejection can coexist with stalled proposals or an omitted scheduled repair. On copied failure witnesses, one-component interventions distinguish these mechanisms from an incorrect collision veto. Conditional arguments explain why confirmation fidelity alone guarantees neither progress nor completion. Native functional coverage of seven teachers and three architectures supplies context under heterogeneous finite protocols; it does not identify a universally correct or uniformly superior learned algorithm.

## 1 Introduction

Two adjacent pending vertices repeatedly propose the same free color. Their learned collision veto correctly rejects both, yet the execution reaches its horizon without finishing. In another learned policy, a scheduled repair proposes an occupied color; a correct rejection leaves the original conflict unrepaired because the schedule does not retry. These are observed frozen transfer failures in our coloring framework. They motivate asking which local rule a policy learned, beyond whether its usual runs end properly.

A second example makes the identification problem sharper. On the observed teacher states, passive, matched and counterfactual BE/E1 procedures each match every measured native action. On the same states with typed proposal interventions, their responses differ dramatically. A successful native choice constrains the accepted option but need not determine what happens to an occupied option or a silent root. Repeated graph testing alone cannot resolve that ambiguity.

We make three concrete contributions. First, we specify continuation targets for seven implemented teachers, locating interventions between two learned communications and accounting for the supplied interpreter. Second, we reconstruct a paired continuation diagnostic, expose its class imbalance and local-type overlap, and add a separate prospective comparison with equal supervised-target budgets and held-out sufficient COMMIT input types. We quantify what that split withholds, class coverage and programmed diagnostic controls; it is not a test of new logical conditions. Third, we connect frozen proposal/repair failures to their missing progress hypotheses using one-component interventions on preserved witness executions. These are claims about specified finite experiments and interfaces, not a new fastest coloring algorithm, general learning theorem, or architecture ranking.

**Relation to previous work.** Neural graph coloring predates this study (Lemos et al., 2019), as do graph-algorithm execution with MAX aggregation (Veličković et al., 2020) and trace-based reasoning benchmarks (Veličković et al., 2022; Veličković and Blundell, 2021). DAgger addresses learner-induced state distributions (Ross et al., 2011); causal confusion in imitation already motivates interventions (de Haan et al., 2019). Our fixed-bank teacher continuations are not a new formulation of these general ideas. GNN/local-computation connections and information restrictions are established (Loukas, 2020; Sato et al., 2019). Alignment and schedules affect extrapolation (Xu et al., 2021; Mathys et al., 2023); asynchronous robustness requires its own assumptions (Solodova et al., 2025), which we do not test. Recent coloring methods learn parallel orderings (Langedal and Manne, 2025), use recursive warm starts (Vanderbush and Weber, 2026), or learn contrastive coloring geometry with transfer (Le et al., 2026). Recent sufficient conditions for provable algorithm learning (Wittig et al., 2026) go beyond our finite extrapolation evidence. Our specific increment is the combination of scoped teacher interfaces, paired continuation diagnostics and replayable failure mechanisms, rather than neural coloring or size transfer themselves. Hint-ReLIC already regularizes algorithmic predictions across input augmentations preserving the next trace step (Bevilacqua et al., 2023); we instead acquire explicit teacher-continuation labels under alternative messages, whose targets can change. No-hint reasoning already separates terminal learning from trajectory imitation (Rodionov and Prokhorenkova, 2023). We claim neither algorithmic intervention nor that distinction as new. Appendix J compares candidate contributions with precedents.

## 2 Interfaces and Continuation Targets

We use finite simple undirected labeled graphs $G = (V, E)$, $n = |V|$, $d(v) = |N(v)|$ and maximum degree $\Delta$. Native training uses $\Delta \leq 3$ and $C_v = \{0, 1, 2, 3\}$. Transfer uses either the global list $C_v = \{0, \ldots, \Delta\}$ or local initial segments $C_v = \{0, \ldots, d(v)\}$. These are not arbitrary list-coloring instances. Partial colors satisfy $x_v \in \{-1\} \cup C_v$, where $-1$ is pending. Availability is $\mathcal{A}_v = C_v \setminus \{x_u : u \in N(v), x_u \neq -1\}$.

A teacher is a specified state machine with observations, private inputs, phases and an interpreter. The learned policy receives declared observations and emits decisions; the interpreter applies them. Anonymous trials use two neighbor exchanges; ColorTrials communicates availability-dependent preferences in the second. JP and Greedy need one exchange mathematically, but their neural decomposition retains two. GM has supplied blocks, barriers and a serial repair queue; BE supplies algebraic predicates and a reduction clock. These are distinct computational regimes, not equal-cost anonymous LOCAL algorithms. Message length and palette computation grow with palette size; we make no CONGEST complexity claim. Recorded native steps count state-machine applications. BE has two point-reduction steps and thirteen KW steps; neural point decisions use the first exchange, whereas KW evaluates the second stage as well. Fifteen native steps are not a fifteen-round learned communication complexity bound.

Mathematical random statements use fresh independent ideal draws. Implementation uses namespaced PCG64 streams and float32 observations. Required IDs are fixed distinct integers in $\{0, \ldots, 65535\}$. The implementation restricts $1 \leq n \leq 65536$, including anonymous experiments for convenience; this is not an anonymous-algorithm size lower bound.

**Evaluation objects.** Functional success means a complete proper in-list coloring within the protocol horizon and satisfaction of checked native invariants. A cutoff is a benchmark failure, not proof of nontermination. Action fidelity compares common states/private inputs. Transition fidelity compares updated states; different action codes need not yield different transitions. Continuation fidelity evaluates a declared oracle $O_T(S, a)$ after a typed message or color-query intervention. Such messages need not be sampled by the native teacher. Execution costs use teacher-specific units. Distributional fidelity concerns the full terminal vector and status on a fixed labeled graph. Terminal validity alone does not imply any of the other properties.

**A continuation is a specified target, not an inferred intention.** Fix a legal teacher-prefix state $S$, including its phase, permissions $w$, IDs, slots, private inputs and derived algebraic data. After the first exchange, a rooted query supplies typed messages $(p_u, c_u)$ at the root and its neighbors: $p_u \in \{0, 1\}$, $p_u \leq w_u$, $c_u \in C_u$ (four option indices for KW). Let $F_v(c)$ mean that the queried color, or the decoded receiver-relative

KW option, is unoccupied. Define

$$E_v = \begin{cases} w_v\,\mathbf{1}[\text{no pending neighbor has a larger ID}], & T = \text{JP}, \\ w_v\,\mathbf{1}[a_v \geq q], & T = \text{BE in KW}, \\ w_v, & \text{otherwise.} \end{cases}$$

The binary confirmation oracle is

$$O_T(S, p, c) = E_v p_v F_v(c_v)\, R_T(S, p, c), \tag{1}$$

where $R_T$ is the rule-specific condition below. It returns a confirmation bit; it does not select a new color, repair an invalid proposal, or promise progress. Its agreement with the native decision is checked when $p, c$ are the native messages. Off-policy targets are defined by this extension of the rule, rather than uniquely identified by the original native trace.

| Rule | $R_T$ and separate targets | Legal state and permitted intervention |
|---|---|---|
| Luby, Johansson | No active neighbor offers $c_v$ | Proper partial state; replace typed participation/candidate messages after first exchange |
| JP | No active neighbor offers $c_v$ | Proper partial state; fixed coherent distinct IDs. New ID assignments precede replay, not independent edge edits |
| ColorTrials | Root preference for $c_v$ strictly exceeds every writable neighbor's preference when that color is free there | Proper partial state; preferences and their availability-dependent messages stay coherent. Silencing $p_u$ does not remove that neighbor's preference message |
| BE | $R_T = 1$ in KW. LINIAL has a separate legal-point target | Proper auxiliary coloring from validated reduction prefix; change option query, not polynomial evaluations, equalities, clock or bounds |
| GM, Greedy | $R_T = 1$ in color phases. GM DETECT separately marks the smaller-ID endpoint of each equal-color/equal-slot edge | Scheduled prefix; GM allows legal temporary same-slot conflicts. Repair queue follows marks. Replace P/C without independently editing marks or queue |

Table 1: Self-contained continuation contracts. The main experiments use rooted message queries, not arbitrary edited global states. Alternative prefixes must satisfy the teacher's native safety/schedule constraints.

For ColorTrials the competing preference is $-1$ when the neighbor is not writable or its queried color is occupied; strict comparison includes ties as rejection. For GM, $F_v$ is computed against actual speculative colors, including during repair. DETECT does not use a P/C continuation label. For BE, a queried index $c$ decodes to $(B - 1 - a_v)4 + c$ at a high auxiliary color, with $q = \lceil 4B/5 \rceil$. Low colors cannot recolor. LINIAL predicts an unblocked evaluation point; the interpreter computes the corresponding polynomial value. This supplied arithmetic is not learned.

**Worked KW continuation.** A replayed training witness has $B = 7$, $q = 6$, root auxiliary color $a_v = 6$ and neighbor auxiliary colors $3, 0, 1$. Its four decoded options are $0, 1, 2, 3$, so only option index 2 is free. With $w_v = p_v = 1$, $O_{\text{BE}}(S, p, c_v = 2) = 1$ and $O_{\text{BE}}(S, p, c_v = 1) = 0$; setting $p_v = 0$ also gives zero. The state, clock and neighbor equalities are identical in the three queries. Native imitation sees the successful choice 2; it need not constrain rejection of choice 1 or silence. The source is seed 94101, root 0, native step 12, tape 14094519662940050106; its full legal prefix is stored with the original acquisition witnesses.

# 3 Teachers and Learned Communication

Names identify the implemented variants. Bibliographic names identify source families, not guarantees imported without checking the variant. Appendix D specifies rules and conditional guarantees.

| Teacher | Exact rule | Supplied information/computation |
|---|---|---|
| Luby | Participate with probability 1/2, propose uniformly available, veto equal active offers | Pending/fixed colors, two private draws; anonymous |
| Johansson variant | Always active, uniform available proposal, equal-offer veto | Anonymous observations; no imported triangle-free theorem |
| ColorTrials | Minimum strictly winning available color, else wait | Four private preferences in $\{0, \ldots, 3\}$ natively; two exchanges |
| JP | Pending strict ID maximum writes first-fit | Coherent relative ID comparisons; fixed IDs |
| BE_BASE_D3 | Two polynomial reductions, thirteen KW reductions | Bounded IDs, fields $13, 7$, point/option equalities, bound/target and phase interpreter |
| Speculative GM | Four-block first-fit, same-slot marks, serial repair | IDs, slots, permissions, phases, queue built from predicted marks |
| Greedy | First-fit in increasing-ID permutation | One externally scheduled write permission per step |

Table 2: Native teachers and distinct information regimes. ColorTrials is the repeated strict-preference subroutine; GM temporary conflicts are legal.

**Shared communication.** E1 NORMALIZED uses node/color MLPs, learned availability, private-input participation, a normalized ordered palette-prefix scorer and learned COMMIT after a second MAX exchange. DeepSets replaces mainly the palette scorer by shared color features and mean palette context. LocalPalette-Transformer uses two-head attention inside each node's palette. Both retain learned graph communication. Functional palette positions are supplied; color-renaming equivariance is not assumed. Hidden/message widths are 64/32. Canonical E1/DeepSets/Transformer have 16,035/17,907/16,995 parameters. Native residual adapters give 19,396/21,268/20,356 where required.

Learned components include participation, proposal scores and confirmation, BE point/option decisions, and GM marks. Polynomial evaluation, schedule construction, decoding, list-membership masks and fixed-order argmax are supplied. There is no exact availability mask, winning flag, eligibility decision or collision veto in neural inference. GM's queue follows predicted marks; an exact teacher never repairs the student's rollout.

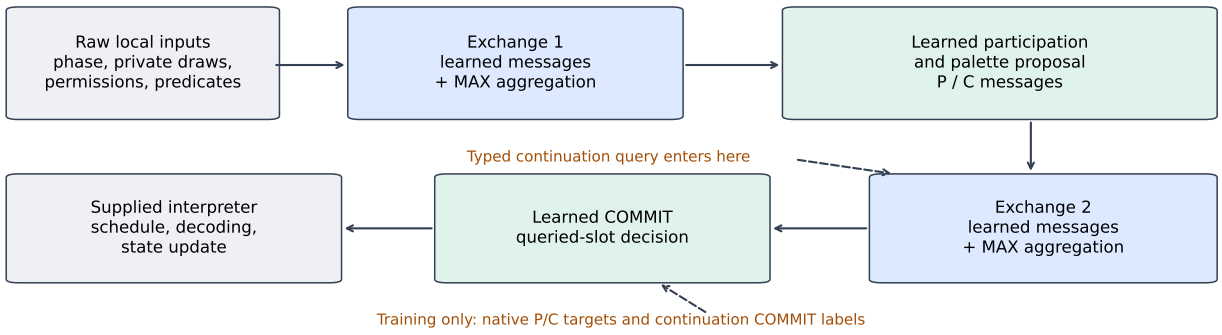



Figure 1: Two learned neighbor exchanges inside a supplied state machine. Each color slot shares parameters; graph messages use MAX. Dashed arrows locate supplied training targets and diagnostic queries. No continuation oracle or exact availability filter participates in an autonomous learned rollout.

The first exchange consumes raw pending/own-color predicates and declared phase, permission, relative-order or algebraic-equality context. Learned features feed participation and palette scores. The second exchange consumes typed P/C messages and raw color predicates, then reads the queried slot's COMMIT logit. ColorTrials instead exchanges per-color preference/features and selects its candidate after that exchange. GM DETECT queries the current color to predict a mark. Hidden-state transformations and message functions

are learned; the interpreter supplies scheduling, list positions, action decoding, program color/auxiliary updates and phase/queue advancement. This decomposition locates the continuation query precisely and separates learned decisions from the externally provided computation.

**Variable-degree representation.** The historical serializer fixes four colors and three neighbor positions. Frozen transfer evaluates the same shared tensors with a sparse structural adapter. Native parity controls cover 21,630 decisions for 150 extensible selected/control checkpoints: no hard-action differences were observed, maximum absolute logit discrepancy $1.526 \times 10^{-5}$. Separately, 1,170 native executions cover all 195 original selected/control checkpoints. These are finite checks, not bitwise-equivalence proofs for every kernel. ColorTrials uses preference/$\Delta$ in transfer, agreeing with preference/3 natively; the out-of-domain scaling choice matters. For native-context teachers, participation averages first-stage features over each receiver's valid palette slots in the structural adapter. Fixed tensors therefore still use an explicitly adapted computation outside the historical serializer; palette membership is not the only adaptation.

# 4 Supervision and Experimental Comparisons

Native training samples sizes 8, 12, 16, 24 at degree at most three and global $K = 4$. Historical banks use six graph families; new banks add edgeless, mixed-component and clique/isolate examples. States, observations and optimization are part of each cohort's identity.

Let $P, C$ denote participation/candidate messages. Passive training supervises native components on teacher prefixes. Its COMMIT loss evaluates the model's current messages against a stored native label, which may be inconsistent with the correct continuation for those messages. Matched training evaluates stored teacher $P, C$ with their concordant native label, without acquiring new labels. Counterfactual training uses the same native bank plus four stored COMMIT labels per color root: it cycles the queried root through four colors with teacher-specific message interventions and a silence case. After matched warmup, its loss is

$$L = L_P + L_C + \tfrac{1}{2} L_{\text{native,matched}} + \tfrac{1}{2} L_{\text{continuation}}.$$

LINIAL point and GM DETECT phases retain their native losses. These fixed teacher-state banks differ from collecting states visited by the learner.

Paired arms share architecture, seed, initialization, native data, minibatch order, Adam learning rate $10^{-3}$ and batch size 64; at a fixed milestone they share updates. CF adds labels and changes the confirmation loss distribution. Thus matched versus CF does not identify an intervention-only effect or efficiency at equal total label budget. Acquired component labels, model access, presentations, updates, bytes and CPU time are separate costs. Replaying a stored label is optimization, not new supervision. Shared banks are charged once for physical acquisition, separately for each model's access.

**Admissibility.** Anonymous queries require proper partial prefixes. JP signs derive from a coherent fixed ID order. BE states come from legal polynomial/KW replay; derived predicates cannot be edited independently. GM requires legal speculative history and schedule. A typed rooted continuation need not form a jointly consistent graph-wide intervention. We claim neither arbitrary state robustness nor joint intervention fidelity from root queries.

**Selection and replication.** The historical sample-budget study has seven budgets $10^3$–$10^6$, 500 updates and five seeds: 735 policies. New procedures record 500, 1,500 and 4,000-update milestones. A prospective amendment expands development from 60 to 300 runs before new confirmation. Five candidate hashes and a final namespace are frozen before a once-only 1,000-run battery per policy. Failed candidates remain; subsequent follow-ups use new namespaces. Local registration is supported by Git/artifact provenance, not independently timestamped external preregistration. Appendix F supplies the budget/provenance table and costs.

# 5 Native Agreement and Continuation Fidelity

We reconstruct the existing paired 4,000-update diagnostics on 60 exact teacher trajectories at sizes 28 and 36 per seed. Their graph/tape namespaces are disjoint from training. They belong to the repeatedly used development split, not an untouched final fidelity battery. BE/E1 gives 28,800/28,800 native permitted-root action agreements in every arm and seed, but very different continuation responses: The paired initial hashes,

| Procedure | Labels | Mean acc. (%) | Seed range (%) | Mean BA (%) |
|---|---|---|---|---|
| Passive | 3,000 | 22.6404 | 9.1607–42.3297 | 60.1413 |
| Matched | 3,000 | 9.2686 | 9.0905–9.3970 | 53.2536 |
| Counterfactual | 6,776 | 99.9932 | 99.9780–100.0000 | 99.9965 |
| Always reject | – | 97.0465 | 96.8800–97.2226 | 50.0000 |
| Eligibility only | – | 98.5990 | 98.5146–98.6208 | 99.2781 |

Table 3: BE/E1, 4,000 updates. Each seed has 99,840 continuation queries. Means describe five paired training seeds; 499,200 queries per arm are correlated within states/trajectories, not independent training replicas. BA averages sensitivity and specificity within seed, then over seeds. Eligibility-only accepts iff the root is KW-eligible and participating; these semantic baselines are diagnostics, not trained competitors.

native arrays, visit counts and final minibatch RNG states agree. CF adds 3,776 labels, changes COMMIT contexts and splits its confirmation loss. This supports a difference between the full procedures, not an intervention-only effect, sample efficiency, or universal functional improvement. Only 2.7774–3.1200% of queries are positive. Every arm has zero false negatives; CF has 34 false positives, all on eligible but silent roots. Most of the raw gap concerns rejection of ineligible KW continuations. Among eligible active queries (4,150–4,598 occurrences per seed), CF is 100% correct in all seeds, whereas passive and matched accept every option, including occupied ones: mean accuracy 67.8117%, BA 50%. Thus the effect also includes learning to reject occupied options, not only a redundant eligibility gate. Always rejecting is accurate overall but destroys progress.

**Novelty of the evaluation queries.** Reconstructing the original query RNG reproduces every numerator. Only 91 of 499,200 occurrences exactly match a complete serialized training query. However, BE COMMIT discards many serialized fields. Its queried-slot logit depends on phase, permission, signed KW offset, root participation, and the MAX-set of occupied/participating/offering neighbor types. Between 99.277% and 99.540% of diagnostic occurrences have one of these sufficient input types already in training. The four queried colors and intervention families are not held out. Among the 549, 459, 717, 522, 722 occurrences with unseen sufficient types, CF correctness is 549, 437, 717, 510, 722, versus matched 158, 342, 229, 182, 246. This posthoc restricted result preserves a seed-wise advantage. Its correlated small subsets and development reuse do not support broad new-intervention-distribution generalization. Appendix G defines the sufficient-input key and gives per-seed numerators. These unseen-type subsets include 52,108,90,56,60 positives and both classes in every seed. All three arms accept all positives; CF's advantage is again specificity on negative continuations. The mean unseen-subset BA is 99.1157% CF versus 66.5299% matched and 50% always-reject.

**Equal weight per sufficient local type.** Occurrence-weighted accuracy gives frequent local configurations more weight. We additionally average accuracy within each sufficient COMMIT input type and then over distinct types, separately within each training seed. This measures the tested type set, not a natural input distribution or a new training replica. Reconstruction with the original RNG and 256-root chunks reproduces every historical numerator before this reweighting.

These estimates retain both the strong rejection difference and the limits of the familiar-type diagnostic. Per-seed seen/unseen, eligibility, silence and occupied/free strata, including all confusion counts, accompany the type-level occurrence counts in the revision evidence. The split uses the union of native and stored CF training types in all historical arms, making its novelty criterion common rather than treatment-dependent.

**Other teachers and dependent architecture comparisons.** At the same milestone GM/E1 has mean continuation accuracy 75.3632% for passive/matched versus 100% CF; CT has 96.6570%, 97.2439%, 99.7452%.

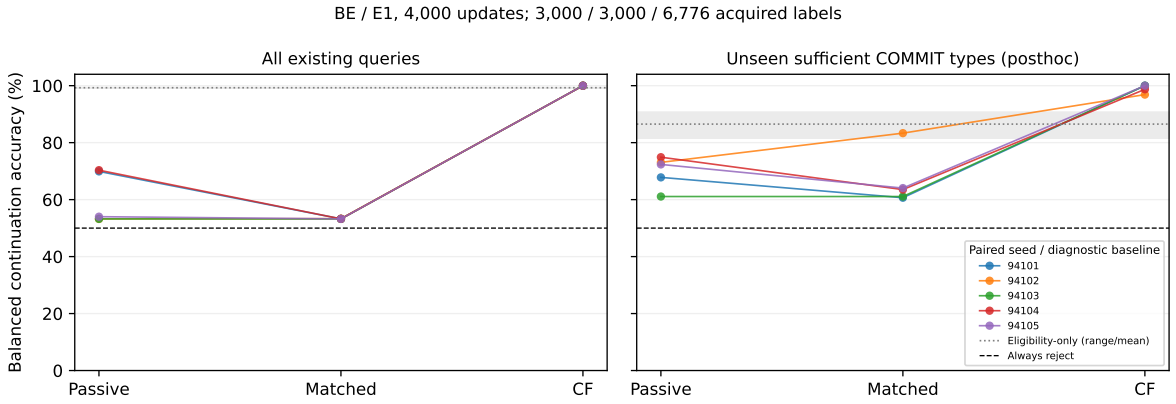



Figure 2: BE/E1 balanced accuracy across paired seeds, with constant-reject and eligibility-only diagnostic baselines. Right: posthoc unseen sufficient COMMIT types. Curves connect treatments within each paired seed; label budgets differ by arm.

| Procedure | Occurrence acc. (%) | BA (%) | Equal-type acc. (%) |
|---|---|---|---|
| passive | 22.640 | 60.141 | 50.784 |
| matched | 9.269 | 53.254 | 46.350 |
| counterfactual | 99.993 | 99.996 | 99.488 |
| always reject | 97.046 | 50.000 | 82.317 |
| eligibility only | 98.599 | 99.278 | 78.973 |

Table 4: Same historical BE queries, three different estimands. Entries average five paired seeds. Equal-type accuracy weights the 115–132 distinct tested types equally within seed, rather than their 99,840 occurrences.

Their denominators vary by seed. These are supporting diagnostics. Paired BE/GM CF COMMIT tensors are identical across the three palette cores at audited milestones: fixed-proposal second-stage losses do not involve their palette scorer. These are five paired confirmation replicas per milestone, not fifteen independent architecture replicas. CT couples second-stage gradients to proposal scoring and lacks that tensor identity.

The scoped answer is positive: this CF procedure better reproduces these typed teacher continuations, including some unseen local types, even when passive native agreement is already perfect. It is not always necessary, universally functionally superior, or shown more label efficient. The historical comparison remains separate from the new budget-matched experiment below.

# 6 Prospective Equal-Budget Continuation Experiment

We test a separate acquisition-strategy contrast with BE/E1 and GM/E1 (19,396 parameters). Ten fresh paired seeds per teacher give forty models. Before training, a fixed SHA256 partition holds out 20% of sufficient queried-slot COMMIT type buckets, common across seeds. Phase, root/context features and distinct neighbor message types determine this key; MAX removes multiplicity. Neither training arm supervises a held-out continuation type. LINIAL and DETECT remain separate native targets. This is a split of input contexts within the same intervention templates. It does not withhold all coarser oracle-relevant configurations (quantified below).

Each pair shares 3,000 native component targets and 3,600 fresh auxiliary rooted states. On each auxiliary state the native arm acquires one native COMMIT target and CF one prespecified intervened COMMIT target. Both queries must pass the training-type filter to retain that state. GM auxiliary quotas are 1,800 ASSIGN and 1,800 REPAIR; BE uses KW only. Each model therefore accesses 6,600 optimization targets. Native P/C conditioning is supplied to both arms equally and charged separately: 7,200 root values plus retained neighbor values. Simulator computation, rejected states and duplicate queries are also recorded; this

does not equate all teacher-computation costs. The root option cycles through native-option offsets 0,1,2,3 modulo four. Participation is the root permission, except that offset three independently silences it with probability 1/4; neighbor messages remain native. The cursor advances before filtering, including discarded roots. Test queries use the same intervention and retain held-out CF types; their phase quotas are 4,000 KW or 2,000 each ASSIGN/REPAIR. This single-query design differs from the historical four-query diagnostic. Appendix A.1 specifies the key, serialization, hash and acquisition order.

Adam, batch 64 and 1,500 updates are fixed before test, with 100 matched warmup updates and the same subsequent native/auxiliary loss weighting. Paired data, initialization, index sequences and first-stage final tensors agree. All forty weights are frozen before generating 4,000 held-out queries per seed/teacher on sizes 28/36 and evaluating 1,000 autonomous runs per model on sizes 32/48/64. All exact references pass. Every final continuation has zero intersection with its training sufficient-type set; both truth classes occur in every seed.

| Teacher | Native BA | CF BA | Paired difference [95% interval] | Successes N / CF |
|---|---|---|---|---|
| BE | 84.64 | 90.41 | +5.77 [-13.73,+25.26] | 9,204 / 7,405 |
| GM | 50.00 | 100.00 | +50.00 [+50.00,+50.00] | 10,000 / 10,000 |

Table 5: New final experiment. BA is a percentage; differences and intervals are percentage points, averaged over ten paired training seeds. Intervals bootstrap paired seeds (10,000 resamples), conditional on this fixed split/protocol; individual queries are not independent replications. Success totals are out of 10,000 autonomous runs per teacher/arm. No checkpoint selection uses these tests.

GM equal-type accuracy is 35.44% native and 100.00% CF; both arms average 26.01 native steps.

GM gains 50 percentage points in every paired seed without changing any autonomous outcome or mean step count. Its degenerate bootstrap interval reflects those identical observed seed differences, not certainty over all future types or training replicas. BE's mean difference is +5.77 points, but the interval spans zero $(-13.73, +25.26)$. BE equal-type accuracy increases from 51.70% to 86.94%, while the two endpoints weight errors differently. CF adds false negatives in two BE seeds and reduces autonomous successes from 9,204 to 7,405; one further seed has identical failures in both arms. Mean step counts over *all* final runs (14.53 native versus 13.43 CF) include early exits on failure. Every successful BE run takes 15 native steps in both arms; the lower overall mean is not a successful-run speedup. These are substantive negative results for a uniform-benefit claim.

**Post-test diagnostic controls.** The primary endpoint and all original outcomes above are retained. The following controls and support audit were added after inspecting the test; they are descriptive checks, not a revised primary analysis. The programmed rule $G_v = E_v p_v$ uses supplied eligibility and root participation and ignores query occupancy. It requires no training and is a diagnostic, not a learned label-efficiency competitor.

| Teacher | Procedure | BA (%) | Equal-type acc. (%) |
|---|---|---|---|
| BE | Native | 84.64 | 51.70 |
| BE | CF | 90.41 | 86.94 |
| BE | Eligibility/participation | 99.71 | 81.27 |
| GM | Native | 50.00 | 35.44 |
| GM | CF | 100.00 | 100.00 |
| GM | Eligibility/participation | 53.03 | 62.34 |

Table 6: Same final queries and sufficient-type key; means over ten seeds. The programmed diagnostic ignores occupancy. BA averages sensitivity and specificity; equal-type accuracy first averages within each distinct type, then across types within seed. The diagnostic is post-test.

For BE, the programmed rule scores 99.71% BA, above CF's 90.41%, whereas CF scores 86.94% equal-type accuracy versus 81.27%. A single negative type accounts for 95.05–96.30% of each 4,000-query bank. Class balancing therefore leaves the negative subtypes highly imbalanced; high occurrence-level BA alone does not identify the occupancy condition. Among eligible active BE queries, native and the programmed rule accept

995/995 free options but reject 0/228 occupied options. CF rejects 228/228 occupied options but accepts only 814/995 free options. These are pooled descriptive counts over ten banks, not independent training replications. Appendix A.2 reports the full confusion strata.

GM's native control receives 4,344–4,354 positive COMMIT labels and zero negative COMMIT labels in ASSIGN/REPAIR per seed, while CF receives 1,038–1,098 negatives and the test has 2,420–2,491 negatives per seed. The native model accepts all test queries, giving BA 50%; the contrast therefore demonstrates learning rejections absent from native color supervision. DETECT has separate positive/negative conflict supervision and is excluded from these color-label counts. This result does not compare CF with another acquisition strategy supplying informative negative confirmations.

**Support of the held-out queries.** There are 17–23 distinct sufficient test types per BE seed and 82–92 per GM seed, despite 4,000 occurrences. For these color oracles $O_T = E_v p_v (1 - o_v)$, with $o_v$ the queried-option occupancy. A coarser configuration (phase, $E_v, p_v, o_v$) is therefore oracle-sufficient. CF training has already seen configurations covering 99.475–100.000% of BE test occurrences and 100% of GM test occurrences. The original fine-key holdout remains disjoint: novelty is defined by contextual combinations rather than by this coarser configuration. There is no separate evaluation restricted to novel logical conditions. These tests do not establish extrapolation to new logical conditions. Per-seed support counts and logical overlap are in Appendix A.2; no split, label or model was changed.

**Two distinct BE failure diagnoses (post-test).** In the two deteriorated CF seeds, 105001 and 105008, COMMIT BA on the 3,600 acquired auxiliary queries is 99.510% and 100.000%, versus 52.756% and 52.094% on the held-out contexts. Both models visited all auxiliary queries. Failure to fit acquired labels alone does not explain these cases: the observed COMMIT gap concerns generalization across contexts, without identifying its optimization cause or an architectural impossibility.

Seed 105010 has a different defect: both arms complete only 204/1,000 runs, despite final COMMIT BA 99.451% native and 99.949% CF. Their shared first stage omits participation on 13/118 eligible test roots. A replay of the original 32-case cohort localizes the first invariant violation in one selected run: at KW step 7, vertex 31 must recolor as the bound drops from 21 to 17, but remains at 17 because participation is false. Its proposed value 12 is free; changing only root participation makes COMMIT accept it. Thus this state exposes an upstream omission, rather than an incorrect veto of the actual silent message. These are selected post-test diagnostics, without retuning, a continued rescue rollout or an estimate of causes across all failed runs (Appendix A.3).

The primary endpoint concerns held-out continuation types; autonomous success is secondary. This finite comparison concerns the specified acquisitions at equal target/context access. It supports neither extrapolation to new logical conditions nor an advantage over alternative informative-negative acquisition strategies. BE's occurrence-level endpoint must be read with its diagnostic baseline and type weighting; GM's control lacks negative color confirmations. No universal CF superiority, minimum sample complexity or all-source teacher efficiency follows. The protocol, forty checkpoints, all seed outcomes, per-component costs and freeze receipts accompany the new results (Appendix A).

# 7 Frozen Transfer and Failure Mechanisms

The domain-wide totals below are retained from the original closure study. Its complete raw delivery is unavailable here, so these totals are not a new independent regeneration. All 225 checkpoint identities have been recovered by SHA256 from earlier public archives. The new interventions below reconstruct two selected first-failure cohorts directly with those original weights.

**Teacher extensions.** The six non-BE rules admit the parameterized definitions in Appendix D. JP's historical cap becomes $n$, preserving its priority rule. GM retains four blocks/detection/serial repair; Greedy retains its schedule. Palette lists extend the domain. CT also changes its preference alphabet to $\{0, \ldots, \Delta\}$; it is a declared parameterized variant, not identical fixed-$K4$ private inputs. BE is excluded because fixed fields and four-option blocks lack a higher-degree existence proof. Changing those constants would define another mathematical variant and interface.

**Confirmatory design.** After a separate pilot, weights, conditions, namespaces and horizons are frozen. All 150,000 exact-reference executions pass before students run. Ninety primary extensible checkpoints receive 1,000 executions at each of five conditions: global $n = 64, \Delta = 3, 4, 10$; local $n = 64, \Delta = 4$; global $n = 256, \Delta = 4$. Sixty native-certified matched/passive controls and thirty pre-existing CT 4,000-update controls receive the four $n = 64$ conditions. The thirty are not assumed certified. Weights stay frozen, inference uses float32 CPU. Each graph has maximum degree exactly requested. Conditions use a uniform mixture of regular, capped random, bipartite, tree, irregular, hub, disconnected, isolate-containing, heterogeneous and clique families. Capped random is not an Erdős–Rényi law. Instances/tapes are shared across policies, teachers and cores. There are 4,000 distinct labeled graphs, 20,000 tape seeds and 25,000 case–palette pairs in the 810,000 policy observations. Observation, graph and training-replica counts differ.

Raw-color reconstruction gives **808,413/810,000** correct: 1,559 collisions and 28 cutoffs in fifteen of 810 cells. This is accounting across policies, not the success rate of a mean algorithm. Primary policies give **449,791/450,000**: 209 failures in two cells, with 90,000/90,000 degree-three controls. The other 448 primary cells have zero observed failures. Secondary controls give 358,622/360,000. Pointwise checkpoint intervals describe the declared mixture, not 1,000 executions per graph family. Table 7 preserves seed ranges.

| Teacher | Core | $64, 3, G$ | $64, 4, G$ | $64, 10, G$ | $64, 4, L$ | $256, 4, G$ |
|---|---|---|---|---|---|---|
| Luby | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Luby | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Luby | Transformer | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Johansson | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Johansson | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Johansson | Transformer | 5/5 | 5/5 | 4/5 (972–1000) | 5/5 | 5/5 |
| ColorTrials | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| ColorTrials | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| ColorTrials | Transformer | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| JP | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| JP | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| JP | Transformer | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| GM | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| GM | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| GM | Transformer | 5/5 | 5/5 | 5/5 | 4/5 (819–1000) | 5/5 |
| Greedy | E1 | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Greedy | DeepSets | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |
| Greedy | Transformer | 5/5 | 5/5 | 5/5 | 5/5 | 5/5 |

Table 7: Primary checkpoints with 1,000/1,000 successes out of five. Nonperfect cells also give the per-checkpoint success range. Columns are $(n, \Delta,$ palette), G global and L local. Sixteen of eighteen combinations pass all selected conditions.

**Failure mechanisms.** The original study reports replays of the complete inference batch containing the first failure in each failed cell. All fifteen reported replays reproduce final colors, status and steps, preserving the original batch context. Johansson/Transformer 83101 has 28/1,000 degree-ten cutoffs, all on cliques. The first leaves two adjacent pending vertices with free colors $\{0, 2\}$. Both repeatedly propose color 2, correctly vetoed by COMMIT. In the preserved conditional probe each proposes 2 on all 8,192 new draws. This identifies missing observed proposal diversity in that state, not an incorrect veto or exactly zero mass for the other color.

GM/Transformer 94102 has 181/1,000 local degree-four failures in each of CF, matched and passive. In the first primary CF witness, exact assignment and detection precede a wrong repair proposal: correct rejection omits a scheduled repair and leaves a conflict. Matched/passive first witnesses include a forbidden write. Equal failure counts therefore do not imply identical local mechanisms. A separate exact-prefix control fixes $\Delta = 3$, IDs and endpoint colors but changes a degree-one endpoint's list from four to two; the Transformer misses the exact repair. This isolates palette-context sensitivity in that constructed state, not universal capacity versus optimization explanations.

**One-component interventions on copied witnesses.** We reconstruct each original full inference cohort from its case/tape seeds and recovered checkpoint SHA. The unchanged execution matches its stored terminal colors, steps and status, and all actions match through the selected intervention frame. At that frame only one candidate is replaced by a free alternative; all other messages and first-stage tensors stay fixed and the

learned second exchange/COMMIT is recomputed. A separate control substitutes exact confirmation on the unchanged proposals once. Neither intervention changes a training run or any reported autonomous test.

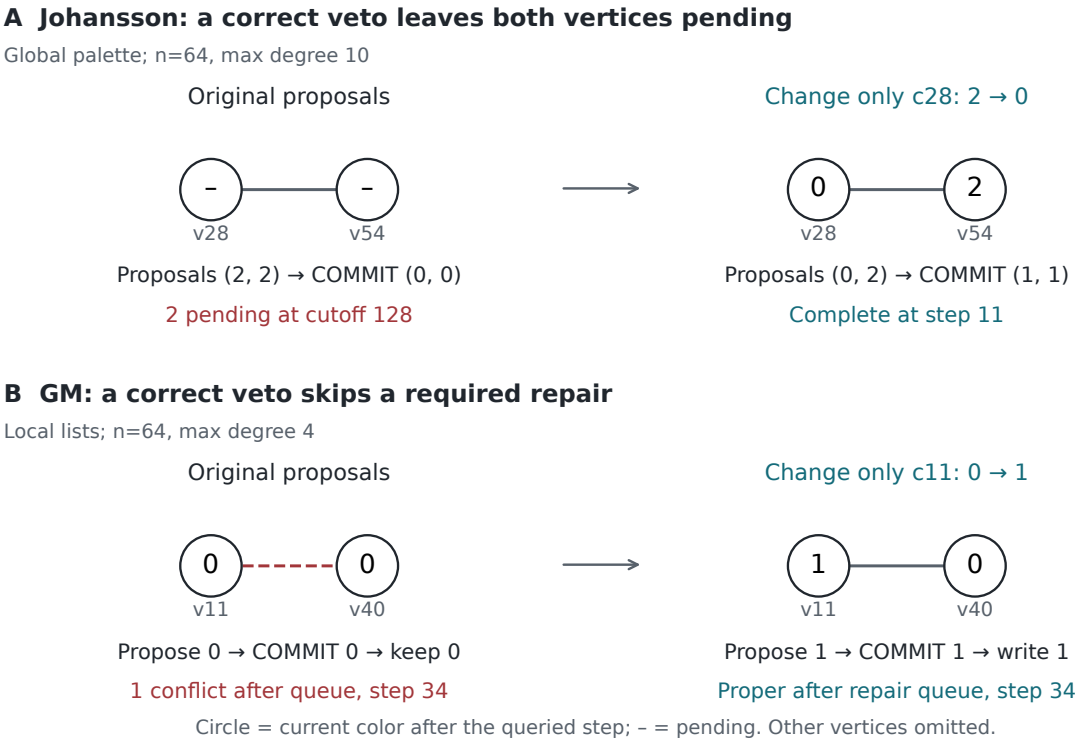



Figure 3: Two selected copied witness executions, showing local fragments after one queried application. A: Johansson steps 10 to 11, with free colors $\{0, 2\}$ at both vertices. B: GM steps 32 to 33, with list $\{0, 1\}$ at the scheduled vertex 11; the dashed edge is an existing conflict. Changing one proposal and recomputing learned COMMIT completes each copied run. Exact-COMMIT substitution on unchanged proposals preserves both failures. All other messages and first-stage tensors stay fixed at the intervention; these cases do not estimate a population rescue rate. Full outcomes are in Table 21.

These controls separate a proposal defect from an incorrect veto in the selected states. Exact-COMMIT substitution is an identity at these already correct vetoes, not an all-state oracle replacement or an audit of COMMIT everywhere. The controls do not establish that changing the proposal module would fix every failed execution, or that the first latent calibration discrepancy caused the observed terminal failure.

Eleven other failed cells involve three matched E1 CT checkpoints, including degree-three failures and incorrect simultaneous confirmations. Higher degree cannot explain all of them. Alpha-threshold discrepancies are latent calibration diagnostics, not supervised availability errors or causes by themselves. The first temporal discrepancy is not automatically sufficient to explain a final failure. Appendix H lists every cell.

**Exploratory scope.** The separate pilot has 3,526/3,570 successes: two collisions, 42 CT resource cutoffs. It spans regular $n = 256$, degrees 2,3,4,5,10,20,50,100,200, sizes through 1,024 at degree 3 or 10, plus heterogeneous/local cases. A ten-second cap was applied only to unfinished dense CT cells after some longer degree-50 successes existed. These unequal resource horizons do not permit homogeneous success-rate ranking. One run per checkpoint/cell is exploratory. The later size pilot gives 240/240 for sixty Luby/Johansson/CT/JP checkpoints on regular $n = 2048, 4096$, degrees 3,10, one run per point. It contains four labeled graphs and twenty tapes, separately from confirmation. GM/Greedy size pilots stop at 1,024; BE stays outside this extension. Confirmation therefore reaches degree ten at size 64, and size 256 at degree four, not their full cross-product, degree 200, or size 4,096.

**Scale witness and confounding.** Degree changes global $K$ and CT preferences simultaneously. A posthoc typed probe keeps an edge endpoint's degree one and local list $\{0, 1\}$ fixed, while a separate star raises global $\Delta$. Preferences $\Delta$ and $\Delta - 1$ make it the strict color-zero winner. All fifteen primary CF CT policies reject its queried COMMIT from degree ten in the main embedding. Legacy preference/3 units alter responses but do not fix all tested cases. Both encodings preserve order. This is scale sensitivity of fixed policies, not an information obstruction, a demonstrated rounding cause or a random functional failure rate. No degree-only causal effect independent of $K$ and private-input scale is identified.

# 8 What Confirmation Can and Cannot Guarantee

Safety and progress require different hypotheses. Exact availability and equal-offer rejection preserve partial properness; always waiting is also safe. For independent fresh proposal laws $q_v$ and participation $p_v$, conditioned on history before those draws, the exact anonymous rule gives

$$\Pr(v \text{ commits} \mid S) = p_v \sum_{c \in \mathcal{A}_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)).$$

Uniform available proposals yield the exact teachers' positive progress bounds (Appendix D). Concentrating both endpoint proposals on the same color can make the product zero under always-active Johansson, even with perfect COMMIT. A finite draw probe demonstrates observed concentration, not a proof of zero alternative probability.

GM uses a different argument: exact marks cover all speculative conflict edges and every queued repair writes a currently free color. The queue visits each marked vertex once. Correctly rejecting an occupied proposal satisfies the continuation rule but violates the required repair-write condition; there is no automatic retry. The two witness interventions test these specific missing hypotheses directly.

The finite-type MAX condition in Appendix E certifies one canonical COMMIT predicate when every required local type satisfies the sign inequalities. It does not certify candidate diversity, scheduled repair, or palette-context invariance. All 45 selected canonical transfer cores fail that sufficient condition; five *different* historical E1 weights pass conditional certificates. Successful finite runs do not convert either cohort into a universally certified learned algorithm. The elementary ID-observation obstruction and representation controls appear in Appendix C; numerical sensitivity is distinguished from missing information.

# 9 Discussion and Limits

Continuation tests identify behavior under a declared extension of a rule; they do not establish that the native trace uniquely specifies that extension. The historical BE comparison shows a procedure difference under unequal budgets and reused development queries. The new experiment isolates an acquisition-strategy contrast at equal target and conditioning budgets on held-out sufficient input types. The class/support audit changes its interpretation: native GM color confirmations are all positive, and the BE occurrence-level metric rewards an eligibility rule that ignores occupancy. CF's stronger BE type accuracy coexists with false negatives and fewer autonomous successes. Nearly all test oracle configurations already occur in CF training. Neither new-condition extrapolation nor an advantage over other informative-negative acquisition strategies is established. Paired first-stage tensors, reused states and graph/tape sharing are recorded rather than counted as independent training replications.

Correct confirmation and autonomous completion are distinct outcomes. The rescues demonstrate sufficiency of a candidate change in two selected witnesses, not a population-wide remedy or a deployable oracle-assisted policy. The selected new BE diagnoses distinguish COMMIT generalization failures despite acquired-label fit from an upstream participation omission with a correct conditional COMMIT response. The latter is localized in one original run; a local participation query does not establish a complete-run rescue. Degree transfer also changes palettes and, for ColorTrials, preference scale; the posthoc scale probe does not identify a degree-only effect. Pilot and confirmatory domains differ; size 4,096 and degree 200 are exploratory limits at different coordinates, not a jointly tested operating region.

The supplied schedules, algebraic predicates and decoding delimit the learned algorithm. Ideal-randomness proofs require explicit conditions not established uniformly for these networks. All-pass native packages use

heterogeneous protocols (Appendix B), and no architecture ranking follows. Output-law studies concern complete vectors/status, not just properness or one-node marginals; nonrejection is not equality (Appendix I). The forty original Pedro weights are available in this workspace and match their inventory hashes; this corrects the earlier local-availability statement but does not constitute a new replay of that cohort. The later closure archive and its 810,000 raw student rows remain unavailable here. Its reported totals are retained with source provenance, not presented as independently regenerated. Independent scientific review of this revision remains pending.

# 10 Conclusion

Perfect measured native agreement can coexist with wrong continuation responses. Auditing the target contract, class balance and local input types makes that gap interpretable. A separate prospective comparison addresses a narrower equal-budget question. Its gains must be interpreted against label coverage, the programmed diagnostic and the limited novelty of held-out contexts. In BE, improved type accuracy does not prevent worse autonomous completion. Frozen failures and component interventions show why correct rejection alone does not supply proposal diversity or a required scheduled repair. These distinctions provide testable explanations of learned local behavior while keeping finite evidence separate from universal guarantees.

# A New Prospective Replicas and Costs

Table 8: All new final seed outcomes. BA is a percentage; differences are percentage points. Successes are out of 1,000 per model. N expands native confirmations.

| Teacher | Seed | N BA | CF BA | Difference | N success | CF success |
|---|---|---|---|---|---|---|
| BE | 105001 | 99.435 | 52.756 | -46.679 | 1000 | 100 |
| BE | 105002 | 99.334 | 99.872 | +0.538 | 1000 | 1000 |
| BE | 105003 | 99.539 | 99.987 | +0.448 | 1000 | 1000 |
| BE | 105004 | 50.193 | 99.923 | +49.730 | 1000 | 1000 |
| BE | 105005 | 50.320 | 99.885 | +49.565 | 1000 | 1000 |
| BE | 105006 | 99.152 | 99.897 | +0.746 | 1000 | 1000 |
| BE | 105007 | 50.348 | 99.923 | +49.575 | 1000 | 1000 |
| BE | 105008 | 99.195 | 52.094 | -47.101 | 1000 | 101 |
| BE | 105009 | 99.462 | 99.859 | +0.397 | 1000 | 1000 |
| BE | 105010 | 99.451 | 99.949 | +0.498 | 204 | 204 |
| GM | 105001 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105002 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105003 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105004 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105005 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105006 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105007 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105008 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105009 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |
| GM | 105010 | 50.000 | 100.000 | +50.000 | 1000 | 1000 |

Each pair physically acquires 10,200 optimization targets: one shared 3,000-target base and two 3,600-target auxiliary banks. Each model accesses 6,600 targets, so shared physical acquisition and summed model access differ. The shared auxiliary P/C context contains 7,200 root scalar occurrences and additional real-neighbor occurrences. Base root P/C context overlaps base target labels; its neighbor conditioning values are also recorded separately. These are scalar occurrences, not claims about distinct simulator calls.

| Teacher | Base target component | Count range across ten banks |
|---|---|---|
| BE | confirmation | 944–944 |
| BE | linial point | 168–168 |
| BE | participation | 944–944 |
| BE | proposal | 944–944 |
| GM | confirmation | 744–754 |
| GM | conflict mark | 738–768 |
| GM | participation | 744–754 |
| GM | proposal | 744–754 |

Table 9: Base supervision sums to exactly 3,000 per seed. Both arms add 3,600 auxiliary COMMIT targets. Context values and simulator costs are reported separately, not absorbed into a free-label assumption.

`results/paper-revision-20261030/prospective-summary.json` records the per-seed component counts, complete-query duplicates, phase quotas, contexts considered/rejected, observed simulator states, acquisition CPU/wall time and paired bootstrap outputs. Training records separately retain presentations, visited targets, source/data hashes and optimizer CPU/wall time. One run's wall time includes a shared-host memory stall; wall time is not a FLOP or teacher-sample-efficiency estimate. All training remains at update 1,500. The original source/config freeze predates scientific training; the complete forty-weight freeze is dated 30 October 2026, 10:11:58 UTC, before any new final diagnostic generation. No final outcome selects a milestone, protocol change, seed exclusion or replacement model.

The recorded source freeze covers the enumerated files, not a complete transitive dependency or environment lock. In particular, graph generation, storage and sample-complexity data helpers outside that list are not sealed by this receipt. Their stability is not independently timestamped here. The original freeze is preserved; this limitation is not repaired by retrospectively adding hashes to it.

## A.1 Prospective acquisition and sufficient key

This specifies the already executed acquisition, not a revised protocol. Color queries use four option indices, $q = c_v \in \{0, 1, 2, 3\}$, degree at most three, and legal teacher-prefix states. Auxiliary and diagnostic graph streams use separate namespaces; their sizes are respectively 8/12/16/24 and 28/36. The frozen configuration and graph generators identify the streams.

For each queried color, let $w_j$ be permission and $u_j$ the supplied pending-or-permitted bit in GM. In BE the queried-stage adapter sets $u_j = 1$. Let $b_j$ be the current-own-color indicator for $q$ in GM and zero in BE. Let $o_{vj}$ be the neighbor's occupancy of the receiver's queried KW option in BE and $b_j$ in GM. Set $\kappa_v = \operatorname{sgn}(a_v - q_{\mathrm{KW}})$ in KW, where $q_{\mathrm{KW}} = \lceil 4B/5 \rceil$, and zero in GM. Here $a_v$ is the auxiliary color; it is distinct from the queried option index $q$. GM also supplies $\sigma_{vj} = \operatorname{sgn}(ID_j - ID_v)$ and the same-slot indicator $\tau_{vj}$; both are zero in BE's queried-stage adapter. With the actual supplied messages $p_j, c_j$, define

$$r_v = (u_v, b_v, p_v, \mathbf{1}[u_v > 1/2]), \qquad m_{vj} = (u_j, o_{vj}, p_j, \mathbf{1}[u_j > 1/2 \ \wedge \ c_j = q], \sigma_{vj}, \tau_{vj}),$$

$$K = (T, \phi, w_v, \kappa_v, r_v, \operatorname{sort}\{m_{vj} : j \in N(v)\}).$$

The neighbor collection is a set: identical message types occur once. This is sufficient for the fixed-weight queried-slot MAX computation, not a minimal oracle quotient, a proposal key or a classifier of LINIAL/DETECT. The original code represents the BE offset as the signed encoded offset; its sign is exactly $\operatorname{sgn}(a_v - q_{\mathrm{KW}})$.

For the hash, all bits are integers 0/1 and signs are $-1, 0, 1$; $T$ is the string `BE_BASE_D3` or `SPECULATIVE_GM`. The phase integer $\phi$ is 5 for KW, 1 for ASSIGN and 3 for REPAIR. Tuples/sets are serialized as JSON lists, with neighbor tuples in lexicographic order and separators `,` and `:` with no spaces. Prefix the UTF-8 JSON by the literal salt `revision-20261030-commit-types-v1`, with no separator. Interpret the first eight SHA256 bytes as a big-endian unsigned integer. Residue zero modulo five denotes held-out types. This reserves 20% of hash buckets; it need not reserve exactly 20% of observed types or occurrences.

Acquisition proceeds as follows.

1. Retain native base targets until the 3,000-component budget is filled. Color-phase base queries must have training keys; LINIAL/DETECT use their separate native targets and do not receive this COMMIT filter.

2. In each auxiliary color-phase batch with an unfilled quota, enumerate candidate roots with a running cursor $i$ before any filtering. Set $k = i \bmod 4$ and $c'_v = (c_v + k) \bmod 4$. Draw an independent uniform value for every considered root. Set $p'_v = w_v$, except that $k = 3$ and a draw below $1/4$ set $p'_v = 0$. All real-neighbor P/C values stay native. Advance the cursor for all considered roots, including rejected roots.

3. Training retains a root only when both $K(S, p, c)$ and $K(S, p', c')$ are training types, subject to the 3,600 KW or 1,800 ASSIGN/1,800 REPAIR quotas. Compute its labels only after this filter. The native arm receives one native confirmation and CF one intervened confirmation on the same state.

4. After freezing all forty models, the diagnostic stream applies the same intervention, but retains a root if its intervened key is held out. The native key is unrestricted at test. Retain exactly 4,000 KW or 2,000 ASSIGN/2,000 REPAIR queries per seed, without filtering on truth labels.

The intervention RNG is seeded by the frozen namespace, teacher, training seed, acquisition kind and the literal `intervention`. Native graph/tape streams have their separate base/auxiliary/diagnostic suffixes. Filtering can change offset and silence frequencies; the pre-filter silence probability is not a claim about the retained distribution. No phase quota balances the positive and negative truth labels within that phase.

## A.2 Post-test coverage and confusion audit

All tables here are descriptive additions after the final test. Labels are recomputed from raw encoded state arrays and agree with stored base, auxiliary and final color confirmations. Fine keys are reconstructed from those arrays; their training/test unions equal the delivered key files and obey the original hash filter. Model predictions are summarized from frozen result records; this audit does not rerun neural inference or train another model.

Table 10: Ranges over ten seeds of base plus auxiliary color-COMMIT targets. LINIAL and DETECT have different targets and are excluded. Minima and maxima in separate columns need not occur at the same seed.

| Teacher | Phase | Arm | Positive labels | Negative labels |
|---|---|---|---|---|
| BE | KW | native | 152–189 | 4,355–4,392 |
| BE | KW | cf | 113–147 | 4,397–4,431 |
| GM | ASSIGN | native | 2,417–2,440 | 0 |
| GM | ASSIGN | cf | 2,108–2,153 | 277–326 |
| GM | REPAIR | native | 1,904–1,932 | 0 |
| GM | REPAIR | cf | 1,119–1,171 | 749–793 |

Table 11: Final support and programmed diagnostic by seed. Types are distinct fine sufficient keys; mass is the largest key's fraction of 4,000 queries. Overlap is the fraction of occurrences whose coarser oracle configuration was seen by CF training. Mass, overlap, BA and type accuracy are percentages.

| Rule | Seed | Types | Max mass | Overlap | Gate BA | Gate type acc. |
|---|---|---|---|---|---|---|
| BE | 105001 | 20 | 95.38 | 99.750 | 99.679 | 80.000 |
| BE | 105002 | 23 | 96.03 | 99.475 | 99.770 | 78.261 |
| BE | 105003 | 17 | 96.30 | 99.825 | 99.770 | 82.353 |
| BE | 105004 | 20 | 95.17 | 99.750 | 99.614 | 80.000 |

| Rule | Seed | Types | Max mass | Overlap | Gate BA | Gate type acc. |
|---|---|---|---|---|---|---|
| BE | 105005 | 18 | 95.62 | 100.000 | 99.731 | 83.333 |
| BE | 105006 | 20 | 95.05 | 99.800 | 99.589 | 80.000 |
| BE | 105007 | 21 | 95.15 | 99.725 | 99.704 | 80.952 |
| BE | 105008 | 20 | 95.53 | 100.000 | 99.706 | 85.000 |
| BE | 105009 | 22 | 95.67 | 99.725 | 99.756 | 86.364 |
| BE | 105010 | 17 | 96.23 | 99.775 | 99.757 | 76.471 |
| GM | 105001 | 89 | 23.80 | 100.000 | 53.055 | 61.798 |
| GM | 105002 | 84 | 23.62 | 100.000 | 52.910 | 63.095 |
| GM | 105003 | 92 | 23.82 | 100.000 | 53.204 | 63.043 |
| GM | 105004 | 86 | 24.60 | 100.000 | 53.182 | 62.791 |
| GM | 105005 | 91 | 23.88 | 100.000 | 52.686 | 63.736 |
| GM | 105006 | 83 | 24.10 | 100.000 | 52.921 | 60.241 |
| GM | 105007 | 86 | 24.52 | 100.000 | 53.193 | 62.791 |
| GM | 105008 | 87 | 23.85 | 100.000 | 53.111 | 60.920 |
| GM | 105009 | 89 | 23.80 | 100.000 | 53.065 | 64.045 |
| GM | 105010 | 82 | 24.60 | 100.000 | 53.005 | 60.976 |

Table 12: Confusion counts summed over ten final banks, 40,000 queries per teacher/procedure. Within eligible active queries, positives are free options and negatives occupied options. Silent and ineligible queries are negative. No GM query is ineligible. These are accounting totals, not IID replications.

| Rule | Stratum | Procedure | TP | TN | FP | FN |
|---|---|---|---|---|---|---|
| BE | eligible active | native | 995 | 0 | 228 | 0 |
| BE | eligible active | cf | 814 | 228 | 0 | 181 |
| BE | eligible active | gate | 995 | 0 | 228 | 0 |
| BE | eligible silent | native | 0 | 91 | 42 | 0 |
| BE | eligible silent | cf | 0 | 52 | 81 | 0 |
| BE | eligible silent | gate | 0 | 133 | 0 | 0 |
| BE | ineligible | native | 0 | 26,956 | 11,688 | 0 |
| BE | ineligible | cf | 0 | 38,644 | 0 | 0 |
| BE | ineligible | gate | 0 | 38,644 | 0 | 0 |
| GM | eligible active | native | 15,554 | 0 | 22,963 | 0 |
| GM | eligible active | cf | 15,554 | 22,963 | 0 | 0 |
| GM | eligible active | gate | 15,554 | 0 | 22,963 | 0 |
| GM | eligible silent | native | 0 | 0 | 1,483 | 0 |
| GM | eligible silent | cf | 0 | 1,483 | 0 | 0 |
| GM | eligible silent | gate | 0 | 1,483 | 0 | 0 |

The reproducible analysis and input hashes are `scripts/audit-revision-continuations.py` and `results/paper-revision-20261030/continuation-audit.json`.

## A.3 Selected BE failure diagnoses after the test

Frozen-model inference covers both arms of BE seeds 105001, 105002, 105004, 105008 and 105010, plus both GM arms in 105001. This deliberately selected set includes good and bad outcomes; it does not estimate their prevalence. There are 102,128 COMMIT predictions on existing native-color, auxiliary-training and final banks. All 48,000 final predictions reproduce the original confusion counts; no weights, split or primary results are changed. This diagnosis uses neural inference, separately from the preceding array-only coverage audit.

| CF seed | Aux. BA | Final BA | Aux. FN | Final FN | Final FP |
|---|---|---|---|---|---|
| 105001 | 99.510 | 52.756 | 1 | 98 | 10 |
| 105008 | 100.000 | 52.094 | 0 | 83 | 16 |

Table 13: Selected BE COMMIT generalization failures. BA is a percentage. Each row evaluates the arm's 3,600 acquired auxiliary queries and the 4,000 held-out final queries. All auxiliary targets were visited during training; neither selected model has an auxiliary false positive. The contexts differ, so this is a descriptive fit/test contrast, not a new randomized comparison.

The two CF models also classify all 944 native-color base queries correctly. Their held-out-context errors coexist with near-perfect acquired-label fit; this does not identify a unique shortcut, optimization cause or limitation of representational capacity.

For seed 105010, native and CF first-stage tensors are identical. On 4,000 teacher-prefix test states their participation head omits 13 of 118 eligible roots, with no false activation. On 3,600 auxiliary states it omits 4 of 128 eligible roots and falsely activates 1 ineligible root. Auxiliary P/C values condition the COMMIT loss; they are not direct participation-loss targets. These first-stage scores on auxiliary states are evaluations, not evidence of failure to fit a participation training objective there. Candidate options equal the native ones throughout both banks, with no occupied proposals at eligible roots.

The witness is `final-s105010-0000`, a 32-node path in the native arm. Replay preserves the first 32 original cases in their original order, cohort size 32 and root batch size 1,024. All 32 terminal color vectors, step counts, failures, invariant flags and correctness flags match the original records. In the selected case the state before step 7 is a proper auxiliary coloring bounded by 21. Vertex 31 has value 17 and must recolor when the interpreter lowers the bound to 17. Its participation logit is -0.420059, so it waits, leaving 17 outside $[0, 17)$. Its option 0 decodes to the free value 12, also chosen by the exact native rule. The actual COMMIT logit -11.225789 correctly rejects silence. Setting only this root's participation bit to true, with identical first-stage tensors, candidates and all other messages, gives logit +4.629150 and the correct positive oracle label. The observer's re-evaluated actions match the original actions at this frame. The intervention is only a local query; unchanged actions drive the replay. It locates the first violation in this case, without showing that all 796 failed runs share this mechanism or that the intervened run would complete.

`scripts/probe-revision-be-failures.py` reproduces the diagnosis; `results/paper-revision-20261030/be-failure-diagnostics.json` records input hashes, counts, graph/tape seeds and the local intervention. This is post-test explanatory evidence, not independent scientific review.

# B Native Functional Coverage

The native result is **21/21 combinations under their respective protocols**: five selected checkpoints per combination, each with 1,000/1,000 proper complete invariant-valid executions. Twelve (Luby, Johansson, JP, Greedy across three cores) are historical passive packages, seeds 83101–83105. Nine (ColorTrials, BE, GM across three cores) are new CF packages, seeds 94101–94105. The representative BE/CT weights use 4,000 updates and GM 500. Matched controls also close all nine; CF is not shown necessary for native functional success.

Final tests use sizes 32, 48, 64; family weights differ: seven historical strata versus ten new strata of 100 runs. The 105 selected policies account for 105,000 successful observations, not a success probability of an average policy. Five training seeds are the treatment replication units. Under independent Bernoulli trials with a common probability, 1,000/1,000 gives pointwise two-sided 95% Clopper–Pearson lower limit $0.025^{1/1000} = 0.996318$; the one-sided limit is $0.05^{1/1000} = 0.997009$. For fixed independent nonidentical strata, AM–GM makes the zero-failure formula conservative for the weighted average, not each stratum. No simultaneous all-policy guarantee or universal correctness follows. A selected successful budget is not a theoretical minimum. The run-level interpretation also assumes semantically neutral batching. Real-arithmetic message separation supports that premise; universal numerical neutrality across floating-point batch contexts has not been certified.

| Teacher | E1 | DeepSets | Transformer |
|---|---|---|---|
| Luby | H: 5/5 | H: 5/5 | H: 5/5 |
| Johansson | H: 5/5 | H: 5/5 | H: 5/5 |
| ColorTrials | N: 5/5 | N: 5/5 | N: 5/5 |
| JP | H: 5/5 | H: 5/5 | H: 5/5 |
| BE base | N: 5/5 | N: 5/5 | N: 5/5 |
| GM | N: 5/5 | N: 5/5 | N: 5/5 |
| Greedy | H: 5/5 | H: 5/5 | H: 5/5 |

Table 14: Native coverage: each passing checkpoint has 1,000/1,000 successes. H: historical passive; N: new CF. These protocols do not form a homogeneous architecture-ranking experiment.

**Preserved negative results.** The historical study's nine noncertifications under 500 updates are not impossibility results. The pre-revision native-certification study retains 21 failed executions among 195,000 final observations across 39 packages and 170 distinct checkpoints; some weights receive new batteries. All failures remain separate from the selected all-pass set. Independent fixed-bank E1 controls compare 500 versus 1,500 updates along the same seed/checkpoint lineage, holding labels fixed and using fresh final tapes. This is a more narrowly identified optimizer contrast than matched versus CF; it does not establish a CF effect or monotone benefit from more updates (Appendix F).

# C Information and Identifiers

Consider an initial two-node JP graph with identical anonymous observations and fixed private inputs. Swapping the distinct IDs changes the required root action from write to wait without changing the anonymous observation.

**Proposition 1** (Observation indistinguishability)**.** *If two admissible contexts have the same observation and different deterministic target actions, no deterministic function of that observation is correct on both. A randomized function cannot be almost surely correct on both.*

*Proof.* Equal inputs give equal outputs or equal conditional output laws. Different point-mass targets cannot both be realized. For equiprobable binary contexts, average correctness is at most 1/2. □

This elementary interface obstruction is not a new distributed lower bound. It requires indistinguishability, not numerical closeness, and does not preclude solving coloring by a different rule. Historical integrated JP/GM already had numeric ID differences; their failures were not ID blindness.

| JP representation | 94101 | 94102 | 94103 | 94104 | 94105 |
|---|---|---|---|---|---|
| Without ids | 6/60 | 6/60 | 6/60 | 6/60 | 6/60 |
| Numeric ids | 13/60 | 11/60 | 6/60 | 26/60 | 15/60 |
| Numeric differences | 16/60 | 15/60 | 19/60 | 31/60 | 20/60 |
| Relative comparisons | 60/60 | 60/60 | 60/60 | 60/60 | 60/60 |

Table 15: Exploratory E1 JP control: same dimensions, initialization, 3,000 labels and 500 updates. Entries are development successes out of 60, not final 1,000-run certifications.

Relative comparisons achieve 60/60 in all five seeds. This supports representation sensitivity under this budget. Numeric IDs/differences retain order information, so finite failures do not prove inability to learn it. Signs remove the need to resolve gaps as small as 1/65536; they must arise from one coherent order. BE uses actual ID arithmetic: increasing numeric transformations need not preserve its polynomial trajectory.

# D Exact Rules and Conditional Guarantees

Assume finite simple graphs, proper partial states where required, declared lists of size at least $d(v)+1$, coherent observations and the specified synchronous exchanges/schedules. Code domain guards and independent scalar tests complement these arguments.

**Lemma 2** (Availability)**.** *For a pending vertex,* $|\mathcal{A}_v| \geq |C_v| - (d(v) - d_U(v)) \geq d_U(v) + 1$.

*Proof.* At most one distinct color per fixed neighbor is forbidden. Counting duplicates only weakens the bound; differing local lists cause no problem. □

**Luby/Johansson.** Proposals avoid fixed colors, and exact equal-active-offer rejection prevents simultaneous collisions. Permanent colors persist, giving safety by induction. Condition on history/state before new private inputs, not a snapshot already containing those inputs. If $q_v(c)$ is the proposal law, $p_v$ its independent participation, with independent pairs across nodes, then

$$\Pr(v \text{ commits} \mid S) = p_v \sum_{c \in \mathcal{A}_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)). \tag{2}$$

For uniform proposals and $p_u = 1/2$, the chord bound $\log(1-x) \geq -2\log(2)x$ on $[0, 1/2]$ and Jensen yield $\frac{1}{2}2^{-d_U(v)/|\mathcal{A}_v|} \geq 1/4$. For always-active Johansson, each pending neighbor of $v$ has at least two free colors, so $q_u(c) \leq 1/2$. The same argument yields $2^{-2d_U(v)/|\mathcal{A}_v|} \geq 1/4$. These prove the specified variants' bounds; they do not import a different algorithm's theorem (Luby, 1993; Johansson, 1999).

**ColorTrials.** Let $D \geq \Delta$ be the common preference bound: $D = 3$ natively, even on lower-degree graphs, and $D = \Delta$ in the extension. Pending nodes draw independent preferences for each color in $\{0, \ldots, D\}$. Only pending neighbors for which the color is available compete. The minimum strictly winning available color is chosen; otherwise wait. Adjacent vertices cannot both strictly win one color. For any $c \in \mathcal{A}_v$, the event of preference $D$ at $v$ and smaller preferences at all at most $\Delta$ competitors has probability at least $\rho_{\mathrm{CT}} = (D+1)^{-1}(D/(D+1))^{\Delta} > 0$ for $D > 0$. It guarantees some write at $v$, perhaps a lower winning color. Isolates at $\Delta = 0$ finish in one step. The common global preference alphabet matters; different local alphabets need another analysis. This is a repeated subroutine inspired by symmetry breaking (Schneider and Wattenhofer, 2010), not a full multi-phase complexity claim.

**Random termination.** If each pending vertex has conditional progress at least $\rho > 0$ at every reachable state, iterated conditioning gives $\Pr(T_v > t) \leq (1-\rho)^t$ and $\Pr(T > t) \leq \min\{1, n(1-\rho)^t\}$. Thus ideal exact rules terminate almost surely. Transfer uses at least 128 steps, enlarged for CT until this ideal tail bound is at most $10^{-9}$. Neither arbitrary learned policies nor finite PRNG orbits inherit the result automatically.

**JP and Greedy.** JP's pending strict maxima of coherent distinct IDs form an independent set. First-fit exists by availability, and the remaining global maximum writes each step, giving safety and at most $n$ steps. This synchronous variant is inspired by Jones and Plassmann (1993). Greedy permits exactly one vertex in increasing-ID order per step; first-fit avoids prior colors and completes in $n$ writes. Global scheduling cost is supplied, not a free LOCAL operation.

**Speculative GM.** Split increasing-ID order into $\min(4, n)$ balanced blocks. ASSIGN visits one position per block simultaneously. A later assignment avoids all earlier neighbor colors, so only equal-slot vertices can conflict. DETECT marks the smaller-ID endpoint of each equal-color/equal-slot edge. Every conflict is covered; the unmarked subgraph is proper. Serial repair visits marked vertices once in increasing-ID order and avoids all current neighbor colors. The last repaired endpoint of any edge avoids the other; edges with no repaired endpoint were already proper. Correctness follows in $2\lceil n/\min(4,n)\rceil + m$ steps, $m \leq n$. Every scheduled write/mark must be correct. Missed learned repairs are not retried. This is the implemented four-block variant (Gebremedhin and Manne, 2000), including supplied barriers/queue, not a guarantee for arbitrary learned marks.

**BE degree-three base case.** Distinct IDs are initial proper auxiliary colors. Degree-at-most-four polynomials over $\mathbb{F}_{13}$ encode them. A different neighboring polynomial agrees at at most four points, so three neighbors block at most twelve and $13 > 3 \cdot 4$ leaves a point. Encode selected point/value below $13^2 = 169$. A second degree-at-most-two reduction over $\mathbb{F}_7$ has a point because $7 > 3 \cdot 2$, yielding bound 49. Different chosen points give different pairs; equal points have different unblocked values, proving properness (Linial, 1992). For bound $B > 4$ set $q = \lceil 4B/5 \rceil$. Colors below $q$ remain; each high old color owns a disjoint four-option block in $[0, q)$, possible since $4(B - q) \leq q$. Three neighbors forbid at most three options. High–high neighbors have disjoint blocks; high–low collisions are excluded by free options. The sequence $49 \to 40 \to 32 \to 26 \to 21 \to 17 \to 14 \to 12 \to 10 \to 8 \to 7 \to 6 \to 5 \to 4$ has thirteen KW reductions (Kuhn and Wattenhofer, 2006). Decoding does not search for a good point/option. This is only the degree-three base construction associated with Barenboim and Elkin (2008), not the general recursion. At degree four, $13 > 4 \cdot 4$ fails; changing constants changes the variant.

# E Conditional Representation and Probability Results

The following elementary arguments are not claimed new general GNN-learning theorems. Checkpoints satisfy their hypotheses only where verified.

**Proposition 3** (Diffuse proposals)**.** *Assume a proper state, exact rejection, independent node-private draws, independence of participation/candidate within nodes,* $0 < p_{\min} \leq p_v \leq p_{\max} < 1$*, proposal support in* $\mathcal{A}_v$*, and* $\max_c q_v(c) \leq \beta/|\mathcal{A}_v|$*. Then* $\Pr(v \text{ commits} \mid S) \geq p_v(1 - p_{\max})^{\beta d_U(v)/|\mathcal{A}_v|} \geq p_{\min}(1 - p_{\max})^{\beta}$*. Only* $v$ *needs the concentration bound for this individual-node statement.*

*Proof.* The log chord gives $\log(1 - x) \geq ax$ for $a = \log(1 - p_{\max})/p_{\max} < 0$, $0 \leq x \leq p_{\max}$. Write $\Lambda_c = \sum_{u \in U_v} p_u q_u(c)$. Equation (2) is at least $p_v \sum_c q_v(c) e^{a\Lambda_c} \geq p_v e^{a \sum_c q_v(c)\Lambda_c}$ by Jensen. The weighted load is at most $\beta p_{\max} d_U(v)/|\mathcal{A}_v|$; each neighbor has mass at most one on $\mathcal{A}_v$. Multiplication by $a < 0$ reverses that inequality, and availability gives the last bound. Isolates use the empty product. □

**Corollary 4** (Diffuse submass)**.** *If the concentration bound holds only on* $S_v \subseteq \mathcal{A}_v$ *of mass* $\eta_v > 0$*, the lower bound is* $p_v \eta_v (1 - p_{\max})^{\beta d_U(v)/(\eta_v |\mathcal{A}_v|)}$*. Uniform* $\eta_v \geq \eta > 0$ *at every reachable state implies uniform progress* $p_{\min}\eta(1 - p_{\max})^{\beta/\eta}$*.*

*Proof.* Drop nonnegative terms outside $S_v$, renormalize by $\eta_v$, and repeat. $xb^{\beta/x}$ increases for $x > 0$, $0 < b < 1$, since its derivative is $b^{\beta/x}(1 - \beta \log b/x) > 0$. □

**Counterexample 5** (Safety without useful progress)**.** *On a* $K$*-clique with* $K \geq 2$ *colors, deterministic color-zero proposals, participation* $1/2$ *and exact rejection progress only with exactly one active vertex: probability* $K/2^K$*, expected first-progress time* $2^K/K$*. Always-active vertices never progress. A shared uniform candidate draw has uniform individual marginals and the same slow law; independence in* (2) *cannot be replaced by marginal uniformity.*

**Proposition 6** (Canonical finite-type MAX COMMIT)**.** *Assume legal canonical per-color inputs/gating, nonnegative shared second messages, componentwise MAX with zero empty aggregate, and exact proposed-slot gather. The pending active center's queried-slot vector is* $(1, 0, 1, 1)$*; neighbor vectors* $(pending, owns_c, participates, offers_c)$ *are among*

$$(0,0,0,0), (0,1,0,0), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1).$$

*If all 64 type-subset COMMIT signs accept precisely subsets without the fixed-color or active-equal-offer conflict types, the rule is exact at any legal degree/palette in real arithmetic and preserves properness.*

*Proof.* MAX is associative, commutative and idempotent; multiplicities disappear. Nonnegative messages admit zero identity. Gathering the proposed slot removes other slots. The 64 subsets therefore enumerate all aggregates. Excluding the two conflict types prevents old/simultaneous collisions. Induction gives safety, with no progress conclusion. □

This does not cover continuous preference or native residual contexts, multiple offers, SUM or MEAN. Outward-rounded float64 intervals enclose float32-stored weights as real coefficients. A separate mode adds $\gamma_{m+2}$ float32 affine error and a subnormal allowance, conditional on IEEE rounding, no overflow and no TF32/lower precision. It is not a proof for arbitrary GPU kernels. Five historical E1 weights pass; all 45 selected Luby/Johansson/Greedy transfer cores fail the sufficient all-types condition. Failure of a sufficient certificate alone does not prove a reachable failure.

**Proposition 7** (MEAN dilution; SUM construction)**.** *Fix degree/palette-independent messages $m_o, m_f$ and a continuous classifier $F$ for a fixed active center. If $F(m_o) > 0$, a MEAN interface accepts a conflicting star with sufficiently many ordinary neighbors. A SUM interface can represent exact conflict presence in real arithmetic.*

*Proof.* A star with $d$ ordinary leaves and one fixed neighbor owning the candidate has mean $(dm_o + m_f)/(d+1) \to m_o$. Continuity and strict acceptance imply false acceptance at some finite $d$. If $F$ is $L$-Lipschitz, it suffices that $d+1 > L\|m_f - m_o\|_\infty / F(m_o)$. For SUM, use the legal-type conflict bit $b = o + \mathrm{ReLU}(p + a - 1)$ and head $1/2 - \sum_u b_u$. □

The MEAN statement assumes no separate degree/palette input and says nothing about reachability of the constructed state. SUM's observed failure is not impossibility of representing this predicate. An order-free equivariant slot scorer with identical available-slot inputs and invariant pooled context gives tied scores for fixed private input; it cannot select distinct ordered ranks without order features or external tie-breaking. Main scorers have positions, so this obstruction does not explain their errors automatically.

# F Training, Certification and Historical Provenance

Table 16: Selected native packages: actual acquired labels per seed (ranges where necessary), updates and total parameters. Every row has five policies and 1,000/1,000 successes each. Historical seeds 83101–83105, new 94101–94105.

| Teacher | Core | Cohort | Labels | Updates | Parameters | Pass |
|---|---|---|---|---|---|---|
| Luby | E1 | Historical | 999 | 500 | 16035 | 5/5 |
| Luby | DeepSets | Historical | 3000 | 500 | 17907 | 5/5 |
| Luby | Transformer | Historical | 999 | 500 | 16995 | 5/5 |
| Johansson | E1 | Historical | 999 | 500 | 16035 | 5/5 |
| Johansson | DeepSets | Historical | 999 | 500 | 17907 | 5/5 |
| Johansson | Transformer | Historical | 999 | 500 | 16995 | 5/5 |
| ColorTrials | E1 | New CF | 70000 | 4000 | 19396 | 5/5 |
| ColorTrials | DeepSets | New CF | 70000 | 4000 | 21268 | 5/5 |
| ColorTrials | Transformer | New CF | 70000 | 4000 | 20356 | 5/5 |
| JP | E1 | Historical | 999 | 500 | 19396 | 5/5 |
| JP | DeepSets | Historical | 999 | 500 | 21268 | 5/5 |
| JP | Transformer | Historical | 999 | 500 | 20356 | 5/5 |
| BE base | E1 | New CF | 6776 | 4000 | 19396 | 5/5 |
| BE base | DeepSets | New CF | 6776 | 4000 | 21268 | 5/5 |
| BE base | Transformer | New CF | 6776 | 4000 | 20356 | 5/5 |
| GM | E1 | New CF | 6104–6120 | 500 | 19396 | 5/5 |
| GM | DeepSets | New CF | 6104–6120 | 500 | 21268 | 5/5 |
| GM | Transformer | New CF | 6104–6120 | 500 | 20356 | 5/5 |
| Greedy | E1 | Historical | 999 | 500 | 16035 | 5/5 |
| Greedy | DeepSets | Historical | 999 | 500 | 17907 | 5/5 |
| Greedy | Transformer | Historical | 999 | 500 | 16995 | 5/5 |

Historical sources are `results/sample-complexity/certification.json`, frozen candidates and raw final rows. New sources are `results/multiteacher-counterfactual/plans/successful-policies.json` and confirmation freezes/rows. Prefix budgets reuse labels and seeds; they are not independent replications or theoretical sample-complexity scaling. The new packages do not rewrite historical noncertifications as successes.

Source commits include Pedro audit `6836d855`, paper study `b5427faf`, integration `a54a18e3`, sample complexity `61f39760`, counterfactual `291d4b82`, degree `71153993`. These are branches/worktrees in shared history, not independent copies of every experiment. Old rows/protocols remain immutable; the previous manuscript is archived under `paper/ucgnn/archive/pre-closure-71153993.tex/pdf`.

**Optimizer controls and costs.** Independent E1 confirmation compares 500/1,500 updates at nominal 3,000 native labels for Luby, JP and BE, five paired seeds and new final tapes. Only optimizer effort differs along the same training lineage. Six arms and all seed outcomes remain in `results/sample-complexity/optimization-confirmation/results.json`. It identifies additional-optimization effects for those bank/seed conditions, not CF effects or optimal/monotone update counts. Luby and JP complete 1,000/1,000 per seed at both milestones. BE moves from 72,90,84,92,42 successes per 1,000 to 1,000 in every seed; these are the strongest resource-controlled optimizer results in the present evidence. The small-width and no-communication baselines below share native prefixes, 500 updates and paired seed conventions with E1. They are development controls, not optimized competitors or proof that every coloring rule needs communication. No-communication erases received states, relations and masks while retaining local inputs and external permissions. Small uses widths 8/4. Dataset manifests record

| Teacher | E1 | Small | No communication |
|---|---|---|---|
| Luby | 28, 28, 28, 28, 28 | 0, 0, 28, 1, 0 | 0, 0, 0, 0, 0 |
| JP | 28, 28, 28, 28, 28 | 10, 0, 15, 0, 0 | 0, 0, 0, 0, 0 |
| BE base | 8, 5, 5, 6, 3 | 0, 0, 0, 0, 0 | 0, 0, 0, 0, 0 |

Table 17: Native development controls at nominal 3,000 labels and 500 updates. Each entry lists successes out of 28 for seeds 83101–83105 individually.

simulation steps, native/added labels, generation CPU/wall time and bytes. In the historical CF acquisition, four labels per color root are charged even if queries coincide: these are conservative acquisition counts, not distinct input counts. Checkpoints retain Adam, all RNG states, visitation and query caches. The archived closure step itself trained no scientific models; the separate forty-model prospective trial has its own training/freeze receipts. Disposable technical gradient tests produce no weights used in results. The evidence ledger identifies all selected hashes and full label-generation costs.

| Teacher | Native labels | Added CF labels | Bank CPU seconds | Stored MB |
|---|---|---|---|---|
| BE base | 3000 | 3776 | 0.435–0.592 | 0.874–0.876 |
| ColorTrials | 30000 | 40000 | 2.153–2.215 | 1.172–1.186 |
| GM | 3000 | 3104–3120 | 0.857–0.929 | 2.002–2.091 |

Table 18: Acquisition cost of the complete shared native/CF bank, range over five seeds. CPU time includes simulation, encoding and both label types; it is not the isolated marginal CF cost. Decimal MB includes replay witnesses. Training costs are separate.

# G Continuation Diagnostics and Dependence

The sufficient key comes from actual BE second-stage inputs, not its label. Real centers have pending bit one and own-slot bits zero; ID/slot residuals are zero. KW substitutes receiver-relative option equality for neighbor own bits. QueryPolicy signs the offset. The root queried slot has offer bit one; other slots vanish under gather. Each valid neighbor contributes occupied query bit, participation and offer/query equality;

| | Correct / 99,840 | | | | Unseen-type correct | | |
|---|---|---|---|---|---|---|---|
| Seed | Passive | Matched | CF | Subset size | Passive | Matched | CF |
| 94101 | 41,591 | 9,343 | 99,840 | 549 | 229 | 158 | 549 |
| 94102 | 9,157 | 9,322 | 99,818 | 459 | 270 | 342 | 437 |
| 94103 | 9,146 | 9,146 | 99,840 | 717 | 229 | 229 | 717 |
| 94104 | 42,262 | 9,076 | 99,828 | 522 | 288 | 182 | 510 |
| 94105 | 10,865 | 9,382 | 99,840 | 722 | 356 | 246 | 722 |

Table 19: Exact BE/E1 original numerators and posthoc novelty subsets. Subset sizes count correlated query occurrences, not distinct types.

MAX removes duplicates. With phase, root permission, signed offset and root participation, this determines the real queried-slot logit at fixed weights. It is not a sufficient key for proposals. Train keys include native and all stored CF queries. There are 58–65 distinct training keys and 115–132 diagnostic keys per seed, with highly unequal occurrence counts. Complete serialized input hashes retain irrelevant fields and measure a different novelty notion.

The diagnostic reconstruction is `scripts/final-counterfactual-audit.py`. It reuses the original queries/RNG, reproduces all numerators, and records each checkpoint and bank hash. Stored label checks find zero inconsistencies; they validate the stated oracle, not graph-wide realizability of every query. We use five paired seed ranges, not decision-level significance. Even five same-sign pairs give exact two-sided sign-test $p = 2/2^5 = 0.0625$ under an exchangeable-sign null, before multiplicity. Shared BE/GM heads and repeated states cannot increase the independent training sample size. Positive counts across seeds are 3,070,2,895,2,891,2,773,3,115; full confusion/eligibility strata are in `results/final-closure/review-ml-be-strata.json`. For unseen-type subsets positives are 52,108,90,56,60 and negatives 497,351,627,466,662. All arms have zero false negatives. The distinction concerns rejecting negative continuations under the explicitly chosen oracle; native traces alone do not uniquely specify off-policy semantics.

# H Failure Inventory and Statistical Units

The complete table is `results/final-closure/generalization-cells.csv`. The factor table separates teacher, architecture, checkpoint, seed, size, degree, palette, family and cohort within stage. The following lists all failed confirmatory cells, each with 1,000 runs and $n = 64$. `failed-cells.json` and

| Teacher/core | Cohort | Seed | Δ | Successes | Palette |
|---|---|---|---|---|---|
| Johansson/Transformer | Historical primary | 83101 | 10 | 972 | Global |
| GM/Transformer | CF primary | 94102 | 4 | 819 | Local |
| CT/E1 | Native matched | 94101 | 3 | 999 | Global |
| CT/E1 | Native matched | 94101 | 4 | 999 | Global |
| CT/E1 | Native matched | 94101 | 4 | 999 | Local |
| CT/E1 | Native matched | 94105 | 3 | 994 | Global |
| CT/E1 | Native matched | 94105 | 4 | 987 | Global |
| CT/E1 | Native matched | 94105 | 10 | 912 | Global |
| CT/E1 | Native matched | 94105 | 4 | 993 | Local |
| GM/Transformer | Native matched | 94102 | 4 | 819 | Local |
| GM/Transformer | Native passive | 94102 | 4 | 819 | Local |
| CT/E1 | Equal-update matched | 94104 | 3 | 864 | Global |
| CT/E1 | Equal-update matched | 94104 | 4 | 855 | Global |
| CT/E1 | Equal-update matched | 94104 | 10 | 547 | Global |
| CT/E1 | Equal-update matched | 94104 | 4 | 835 | Local |

Table 20: All fifteen failed cells. CT denotes ColorTrials; equal-update controls are not retrospectively assumed native certified.

`failure-replay-audit.json` give full checkpoint/row hashes, graph/tape seeds, first relevant differences and trace paths. All fifteen first failures reproduce with original batch context. Terminal auditing reconstructs graphs/fingerprints, conflicts, incompleteness and palette violations and checks stored invariant/failure fields. It does not pretend to reconstruct unrecorded intermediate states for all runs.

Graphs/tapes repeat across cores, teachers, arms and global/local lists. Each checkpoint's instance-index battery supports its own pointwise inference; pooling dependent observations across policies does not. Realized family counts vary under iid uniform family selection. No aggregate defines a sampled population of trained algorithms.

### H.1 Complete outcomes of the selected component interventions

| Witness | Intervention | Steps | Terminal outcome |
|---|---|---|---|
| Johansson | baseline | 128 | cutoff |
| Johansson | candidate once | 11 | complete/proper |
| Johansson | exact commit once | 128 | cutoff |
| GM | baseline | 34 | collision |
| GM | candidate once | 34 | complete/proper |
| GM | exact commit once | 34 | collision |

Table 21: Posthoc component interventions in two selected first-failure witnesses, not new population success rates. Baselines preserve the original 16-case inference cohort. Exact confirmation is evaluated on the actual learned messages.

The source is `results/paper-revision-20261030/component-rescues.json`. Figure 3 extracts actual vertices, colors and proposals from those records and the original witness frames. The full 64-node graphs, case/tape identities and unchanged 16-case cohort remain part of the diagnosis; the two-node drawing omits the rest of each graph.

## I Output Laws and Preserved Distribution Studies

For fixed labeled $G$, let $\mu_{\pi,G}$ be the full terminal-vector/status law including failures/cutoffs. Labels, palette, interpreter, cutoff, conditioning and color-name quotient must agree before comparing studies. Equal vertex marginals do not determine a joint law: equally weighted $(0,0)/(1,1)$ versus $(0,1)/(1,0)$ have equal marginals and disjoint support. This logical example is not a proper-coloring counterexample on an edge.

**Proposition 8** (Symmetry and coupling)**.** *On a complete graph with exactly $n$ colors, a vertex-equivariant stochastic protocol with exchangeable private inputs is uniform over proper labeled colorings conditional on successful termination of positive probability. Any coupling $(X,Y)$ satisfies $d_{\mathrm{TV}}(\mathcal{L}(X),\mathcal{L}(Y)) \le \Pr(X \neq Y)$.*

*Proof.* Vertex permutations act transitively on the $n!$ proper coloring bijections; equivariance and the invariant success event make their conditional masses equal. For any event $A$, the difference in probabilities of $X, Y \in A$ is at most $\Pr(X \neq Y)$; take the supremum. □

Anonymity alone does not establish the symmetry premises. Untransformed IDs, non-equivariant schedules and numerical ordering require care. With more than $n$ colors, vertex symmetry need not equate different color subsets. For $r$ independent paired runs of one graph/policy protocol with zero disagreements, the one-sided upper limit $1-\alpha^{1/r}$ bounds disagreement and hence TV under that coupling. It says nothing universal about other graphs/checkpoints. Uniform conditional full-transition coupling error $\epsilon$ for $H$ common pre-divergence steps gives $d_{\mathrm{TV}} \le H\epsilon$; per-node error would also require a node union bound. Native accuracy is not this uniform premise.

Global-law, local-palette, robust-statistics and factorial studies remain in their historical directories with schemas, fixed banks and testing families. Complete runs are observations, permutations act on rows rather than independently on vertices, and witness selection needs an independent split or valid joint permutation scheme (Gretton et al., 2012; Hemerik and Goeman, 2018; Phipson and Smyth, 2010; Holm, 1979). Reused

banks are historical reanalysis, not independent new confirmation. Nonrejection does not establish equal laws or tolerance equivalence; power against localized alternatives matters. These studies support scoped distributional distinctions, not new MMD/permutation theory, and are outside the 810,000 denominator.

# J Novelty Comparison and Reproduction Map

| Candidate | Established precedent | Defensible increment |
|---|---|---|
| Neural coloring | Lemos et al. (2019); Langedal and Manne (2025); Vanderbush and Weber (2026); Le et al. (2026) | Seven-rule native coverage with declared information and complete selected-checkpoint provenance |
| Trace imitation; MAX | Veličković et al. (2020, 2022) | Concrete validity/native/continuation comparisons for these teachers |
| Interactive/CF learning | Ross et al. (2011); de Haan et al. (2019); Bevilacqua et al. (2023) | Specific admissibility contracts and paired continuation gaps, not a new general causal method or equal-label advantage |
| Output versus trajectory | Rodionov and Prokhorenkova (2023); Veličković et al. (2022) | Quantified separations in these coloring interfaces, not novelty of the conceptual distinction |
| Information/locality | Linial (1992); Loukas (2020); Sato et al. (2019) | JP representation control and interpreter accounting; the elementary observation lemma is not a new lower bound |
| Algorithm extrapolation | Xu et al. (2021); Mathys et al. (2023); Wittig et al. (2026) | Frozen selected-condition outcomes and replayable palette/preference failures, not a general extrapolation theorem |
| Randomized output laws | Gretton et al. (2012); Hemerik and Goeman (2018); Holm (1979) | Matched statistical objects and protocol-specific evidence, not new test theory |

The focused primary-source search was executed on 26 October 2026, including recent 2026 work. It is not proof of absence of every related study. `docs/NOVELTY_AND_RELATED_WORK.md` records scope and objections. `docs/FINAL_EVIDENCE_AUDIT.md` and `results/final-closure/evidence-register.json` link central claims to proofs, protocols, data, weights, seeds, hashes, limitations and confidence. The preceding records retain their historical scope. The revision's claim matrix and portable reproduction commands are separate. To rebuild its analyses and article from the repository root:

```
PYTHONPATH=src python scripts/reanalyse-paper-be.py
PYTHONPATH=src python scripts/probe-paper-rescues.py
PYTHONPATH=src python scripts/summarize-paper-revision.py
PYTHONPATH=src python scripts/audit-revision-continuations.py
PYTHONPATH=src python scripts/probe-revision-be-failures.py
PYTHONPATH=src python scripts/build-revision-assets.py
PYTHONPATH=src python scripts/build-revised-paper.py
PYTHONPATH=src python scripts/verify-paper-revision.py
```

Checkpoint-based steps require verified public runtimes. The source-only checkout rebuilds the PDF from compact revision analysis outputs. The build receipt checks references and records the actual compiler/environment, PDF text and page renders. Historical closure scripts belong to the archived version; they must not overwrite the revised article or its receipt. The historical closure packet's local path is not a public download. The revision scripts recover published native/CF weights by verified archive and checkpoint hashes, preserve historical result files, and write separate new analysis outputs. The revised build uses native ARM LaTeX with a separate compiler receipt; it does not claim the old x86 Tectonic binary was executed. See `docs/PAPER_REVISION_REPRODUCE.md` for portable commands and

`results/paper-revision-20261030/manifest.json` for delivered file hashes. The pre-revision source/PDF are archived under `paper/ucgnn/archive/pre-revision-c3affd16.tex/pdf`.

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